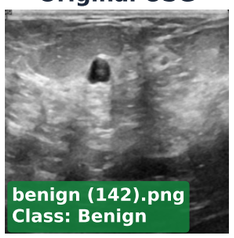
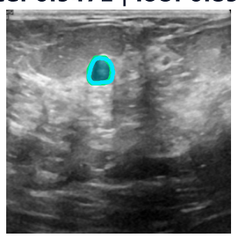
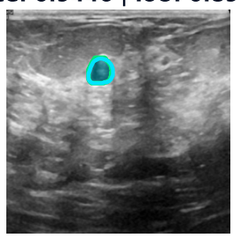
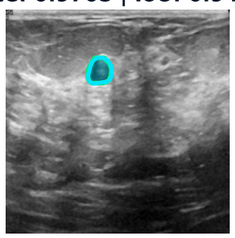
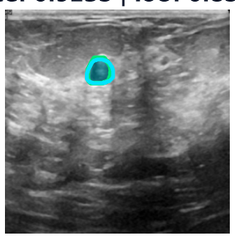
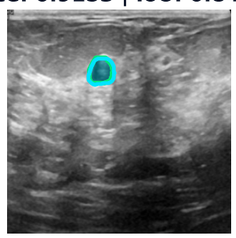
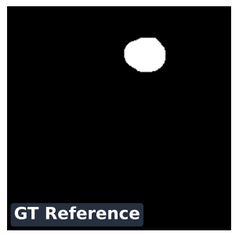
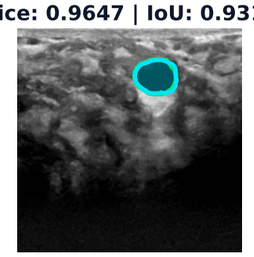
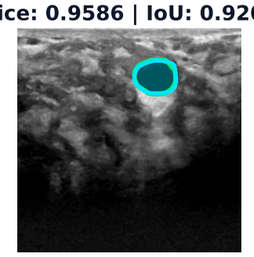
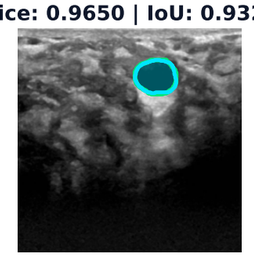
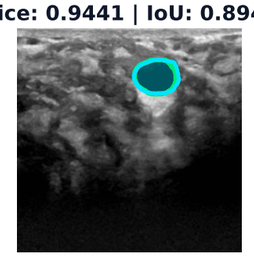
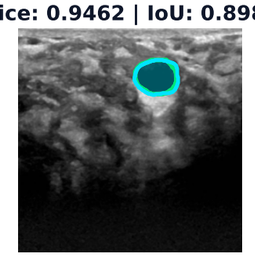
Loss-Function Qualitative Comparison: Attention U-Net Across Five Formulations (Option A)

Comparing Dice, BCE, Combined BCE+Dice (Final Model), Focal, and Tversky Loss Functions on Representative Samples

Ground Truth Boundary (GT) | Predicted Lesion Mask (Model Output) | Predicted Lesion Boundary

1. Ground Truth (GT)	2. Dice Loss	3. BCE Loss	4. BCE + Dice Loss (Final Model)	5. Focal Loss	6. Tversky Loss
					
Original USG	Dice: 0.9471 | IoU: 0.8996	Dice: 0.9446 | IoU: 0.8950	Dice: 0.9705 | IoU: 0.9427	Dice: 0.9235 | IoU: 0.8580	Dice: 0.9153 | IoU: 0.8438
					
					
Original USG	Dice: 0.9647 | IoU: 0.9318	Dice: 0.9586 | IoU: 0.9205	Dice: 0.9650 | IoU: 0.9323	Dice: 0.9441 | IoU: 0.8941	Dice: 0.9462 | IoU: 0.8980
					
					
Original USG	Dice: 0.9024 | IoU: 0.8221	Dice: 0.9189 | IoU: 0.8500	Dice: 0.9445 | IoU: 0.8949	Dice: 0.8535 | IoU: 0.7445	Dice: 0.9337 | IoU: 0.8757
